HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Managed system administration for automation and control systems, ensuring maximum availability and reliability for critical manufacturing operations.
- Executed routine maintenance, patching, and lifecycle management activities for operational technology (OT) infrastructure, adhering to strict GMP regulations and quality standards.
- Developed and maintained comprehensive project documentation, including User Requirements Specifications (URS) and Functional Design Specifications (FDS), for digitalization initiatives.
- Participated in cross-functional project teams, contributing technical expertise to the development and implementation of automation solutions across manufacturing sites.
- Troubleshot complex automation issues, coordinating resolutions with internal engineering teams and external vendors to minimize production downtime.
- Contributed to the standardization of automation systems and engineering processes across global sites, supporting the implementation of Takeda's Global Automation Strategy.
- Containerized Python-based data processing scripts using **Docker** and deployed them to **AWS Lambda**, enabling scalable execution for ETL tasks.

SKILLS

Automation Systems – PLCs, Siemens, Omron, Mitsubishi, HMI, SCADA, iFIX, Wonderware, OPC, Modbus, Ethernet/IP, Profinet, Profibus

Data Management – SQL databases, SQL scripting, Data integrity principles, GAMP, 21 CFR Part 11, EU Annex 11

Networking – TCP/IP, Routing & Switching, IDS / IPS, Active Directory Domain Integration, Firewalls

Software Development – Java, Spring Boot, Python, REST APIs, Docker, AWS (EC2, S3, Lambda, CloudWatch)

Project Management & Compliance – GMP, GDP, URS, FDS, SOPs, Commissioning, Qualification, Change Management, Agile/Scrum

PROJECTS

Manufacturing Data Pipeline Automation (HCL Technologies)

- Developed Python scripts to automate the extraction, transformation, and loading (ETL) of data from disparate enterprise systems, including **PostgreSQL** and **Oracle** databases, into a centralized data warehouse.
- Implemented robust data validation checks and error handling within the Python pipelines, ensuring data integrity and compliance with pharmaceutical industry standards for reporting.
- Scheduled and monitored daily data processing jobs using **Jenkins**, achieving a **98%** success rate for batch data refreshes supporting operational reporting.

Control System Monitoring Dashboard (HCL Technologies)

- Built an operational dashboard using **Java Spring Boot** and **PostgreSQL** to monitor the performance and availability of manufacturing control systems.
- Integrated real-time data feeds from industrial communication protocols (simulated via API) and system logs, providing immediate alerts for critical system deviations.
- Designed and implemented REST APIs for cross-system data exchange, enabling seamless integration with existing IT infrastructure and support tools.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)